

Final Revised Report

A. What Changed From The Old Pipeline

- Old method: single broad LA/AIDS-like block + ad hoc IO realization.
- New method: paper-specific hierarchy with four layers:
 1. Shojaeddini top-tier (QUAIDS/AIDS/LES, leisure-inclusive full consumption),
 2. Jorgenson-Slesnick exact-aggregation benchmark analogue (rank2/rank3),
 3. Döpper-inspired modular diagnostics,
 4. Inforum/LIFT/IdLIFT/Chao bridge + IO propagation architecture.
- Change rationale: improve fidelity to cited methods while preserving feasible identification with local data.

B. Paper-to-Block Mapping

- Shojaeddini et al. (`2021-05.pdf`): top-tier categories, leisure treatment, QUAIDS/AIDS/LES comparisons, ILLS spirit.
- Jorgenson-Slesnick (`Consumption and labor supply ... 2008`): full-consumption + leisure and rank-2/rank-3 exact-aggregation benchmark logic.
- Döpper et al. (`w32739.pdf`): modular heterogeneity-aware diagnostics; no unsupported scanner-data replication claims.
- Inforum/LIFT/IdLIFT/Chao (`LIFT\_Methodology`, `IdLift`, `wp01002`, `wp01004`, `chao\_1991`): bridge matrix D, IO identities, discrepancy-aware documentation, grouped PADS analogue.

C. Formulas Used (Implemented)

- Share-system demand equations (AIDS/QUAIDS analogue):

$$\backslash[\\ w_i = a_i + \sum_j \gamma_{ij} \ln p_j + \beta_i m + \lambda_i m^2 + Z_i' \delta_i + u_i, \text{quad } m = \ln X - \ln P \\ \backslash]$$

- LES comparator (Stone-Geary structural analogue):

$$\backslash[\\ x_i = \gamma_i p_i + \beta_i \left(X - \sum_j \gamma_j p_j \right) \\ \backslash]$$

- Bridge mappings (Inforum/IdLIFT style):

$$\backslash[\\ C = DJ, \text{quad } p_J = D' p_w \\ \backslash]$$

- IO output/price identities (reduced-form):

$$\backslash[\\ q = Aq + f + d, \text{quad } p = A' p + v + \epsilon_p \\ \backslash]$$

- Realized elasticity experiment:

$$\backslash[\\ \text{varepsilon}^{\text{GE}}_{ii} = \frac{\Delta \ln Q_i}{\Delta \ln P_i} \\ \backslash]$$

with $(\Delta \ln P_i)$ generated through bridge-mapped IO propagation from a small producer-side shock.

D. Fully vs Partially Implemented

Fully implemented (within local data limits)

- Shojaeddini-style top-tier QUAIDS/AIDS/LES hierarchy with leisure (approximate ILLS implementation).
- Jorgenson-Slesnick stage-2 rank-2/rank-3 analogue benchmark on full consumption.
- Inforum/IdLIFT-style bridge matrix reconstruction and IO propagation identities in reduced dimension.
- Döpper-inspired modular category diagnostics (reduced-form, heterogeneity-aware).

Partially implemented / not fully identified

- Jorgenson-Slesnick stage-1 intertemporal synthetic-cohort Euler step not identified with 2013-2017 local sample.
- Literal Döpper random-coefficients scanner estimator infeasible without product-store-market scanner data.
- Full 97x92 Inforum bridge unavailable locally; reduced bridge reconstructed from available BEA sector data.
- PADS block implemented in grouped analogue form, not full high-dimensional IdLIFT production configuration.
- Bootstrap percentile interval did not bracket the full-sample QUAIDS estimate; reported as unstable and omitted.

E. Final Preferred Estimates

- Preferred ****E_M**** (top-tier QUAIDS with leisure): ****1.3798****
 - 95% bootstrap CI: not reported due to bootstrap instability in this sample/model.
- Preferred ****E_GE**** (GE-1 top-tier + bridge + IO): ****0.8268****

F. Interpretation

- Magnitude relative to unit elasticity:
 - Preferred Marshallian weighted mean is above unit elasticity in absolute value.
 - Preferred realized GE weighted mean differs materially after bridge/IO propagation.
- Main drivers:
 - Demand-form effects (QUAIDS vs AIDS/LES) and leisure inclusion alter weighted means.
 - Bridge/IO propagation shifts elasticities further via cross-sector price spillovers.
- Dominant category contributions are reported in `top\_tier\_elasticities.csv`, `ge\_elasticities\_top\_tier.csv`, and decomposition notes.
- Old-vs-new numerical comparison is in `old\_vs\_new\_results.csv`.